



Trigeminal Nerve

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DIKIOHS,DUHS.

LEARNING OBJECTIVES

- At the end of the lecture, student should be able to;
- Describe the anatomy of the trigeminal nerve, including its origin, roots, and general functional components.
- Identify and differentiate the three divisions of the trigeminal nerve (ophthalmic, maxillary, and mandibular) with their modalities (sensory/motor).
- Explain the course, major branches, and areas of distribution of each division of the trigeminal nerve.
- Describe the trigeminal sensory system, including:

Trigeminal ganglion

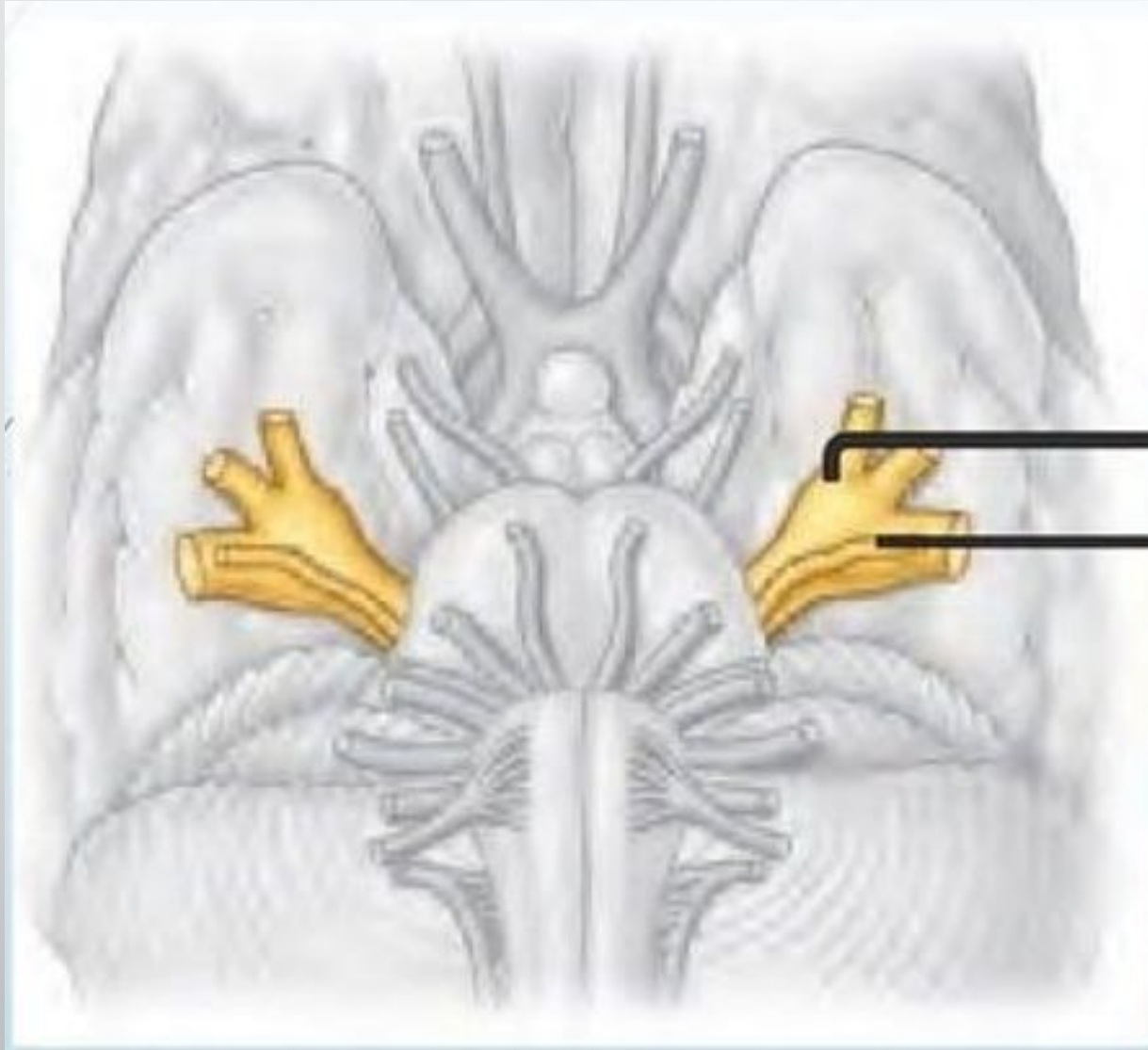
Sensory nuclei (mesencephalic, principal, spinal)

Types of sensations carried

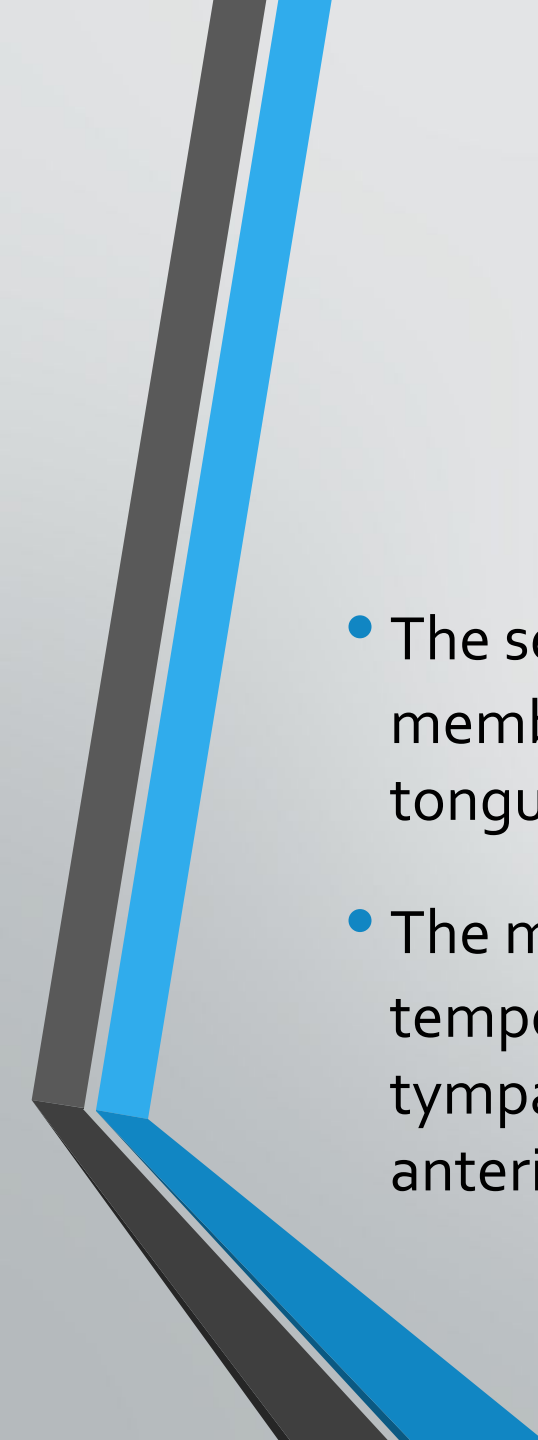
- Correlate the sensory supply of the trigeminal nerve with orofacial structures, including teeth, mucosa, muscles, and skin.
- Understand the role of the trigeminal nerve in orofacial pain mechanisms, including peripheral and central sensitization.

INTRODUCTION

- Trigeminal nerve is the largest cranial nerve. It contains both sensory and motor fibers.
- The name “Trigeminal” was given by Winslow, on account of three divisions.
- It has three branches.
- Ophthalmic_Sensory
- Maxillary_Sensory
- Mandibular_Mixed
- General somatic afferent fibers contains both exteroceptive and proprioceptive impulses.
- It is attached to the lateral part of the pons by its two roots, motor and sensory.

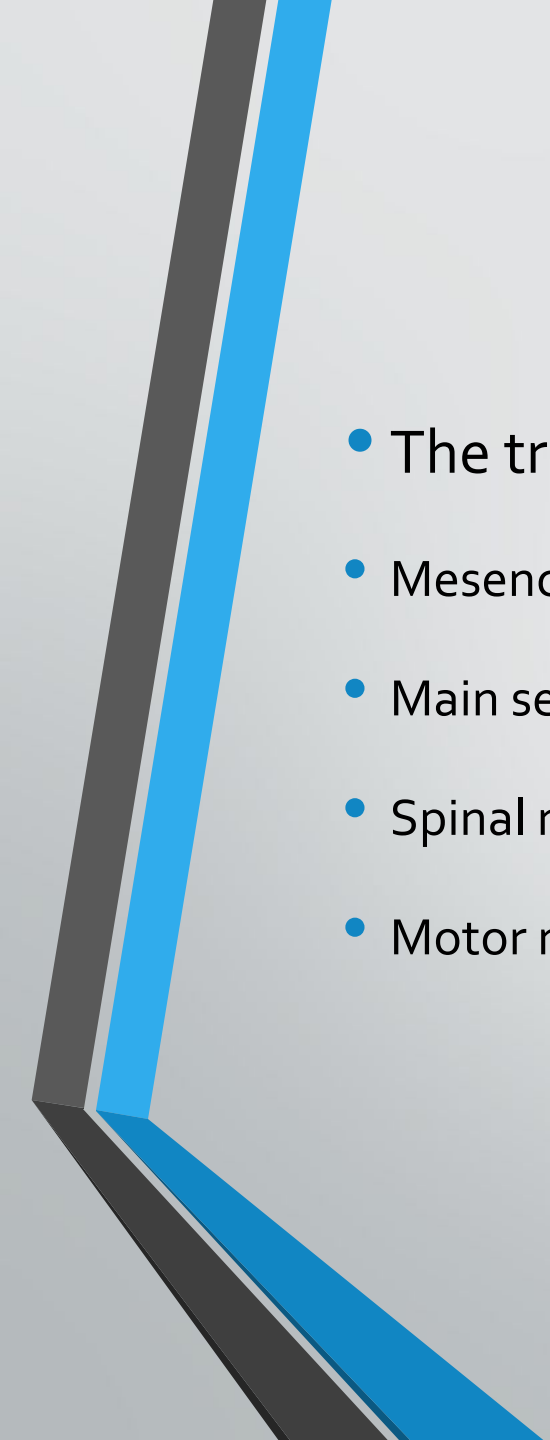


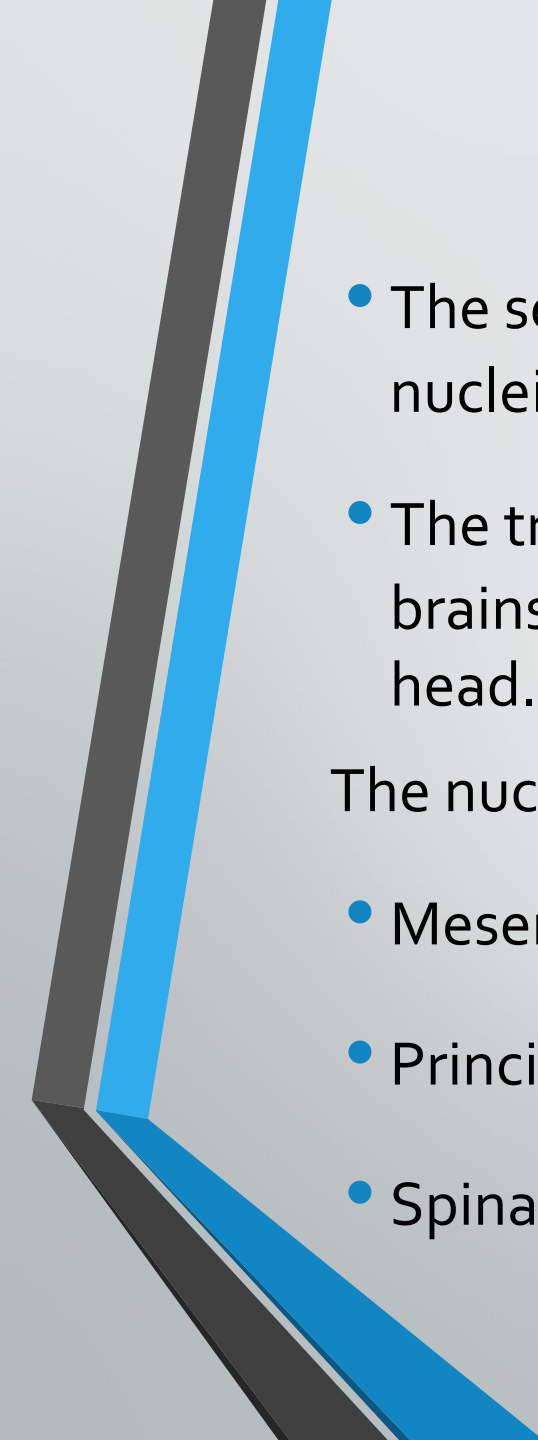
sensor
y
motor

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- The sensory component supplies the skin of the entire face and the mucous membrane of cranial viscera, except for the pharynx and the base of the tongue.
 - The motor component supplies the muscles of mastication (masseter, temporalis, lateral pterygoid and medial pterygoid), muscle of ear (tensor tympani), muscle of soft palate (tensor veli palatini), mylohyoid muscle and anterior belly of digastric.

TRIGEMINAL SENSORY SYSTEM

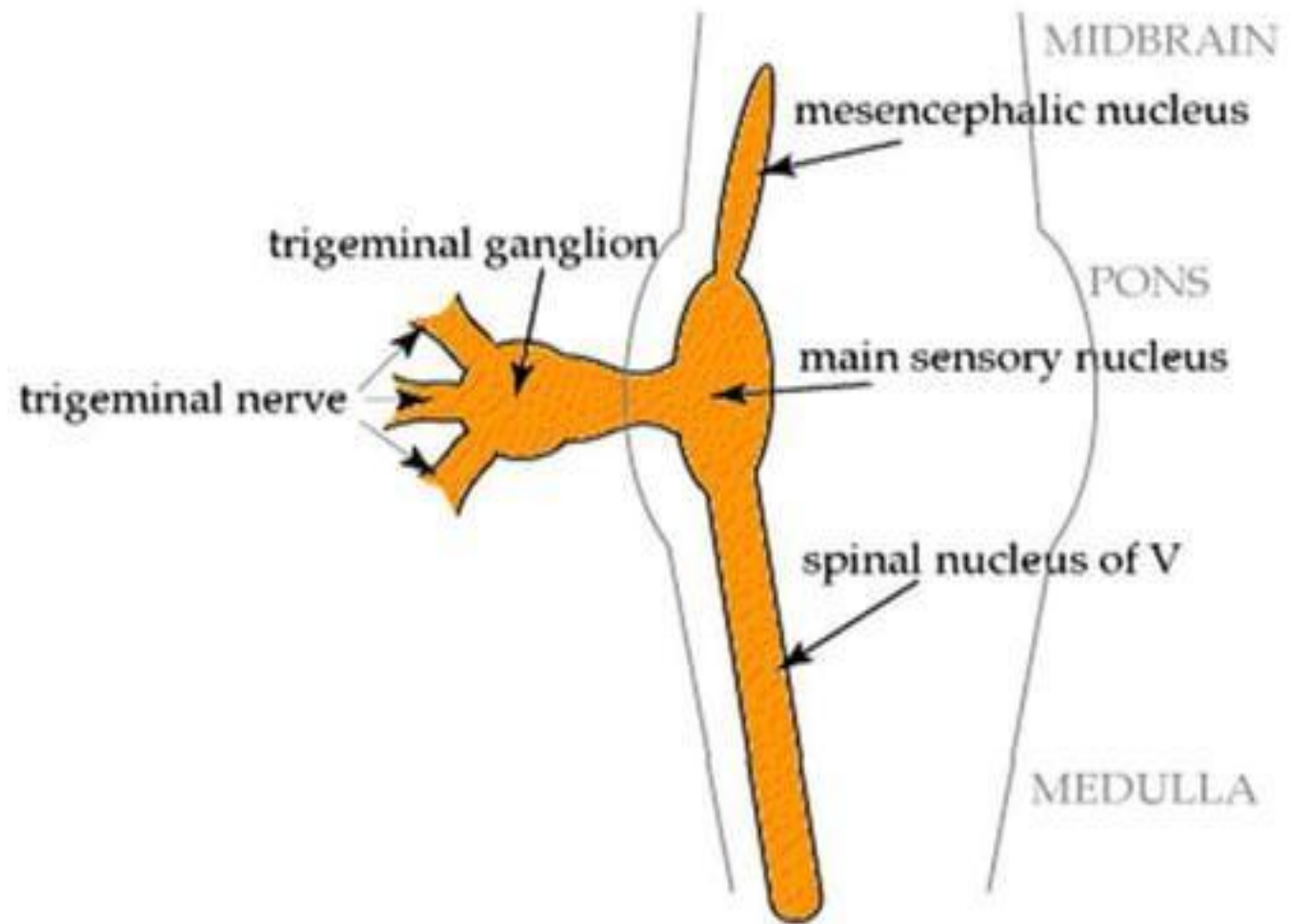
- The trigeminal sensory system consists of the trigeminal nerve, the trigeminal ganglion, and the trigeminal sensory nuclei (the mesencephalic nucleus, the principal nucleus, the spinal trigeminal nucleus, and several smaller nuclei).
- Various sensory signals carried by the trigeminal nerve from the orofacial area travel into the trigeminal sensory system, where they are processed into integrated sensory information that is relayed to higher sensory brain areas.
- Thus, knowledge of the trigeminal sensory system is essential for comprehending orofacial pain.
- This system processes sensory information, including proprioception, touch, temperature, and pain, from the orofacial region, particularly the teeth and their surrounding structures, masticatory muscles, cornea, and the trigeminal nerve territory of the facial skin. In addition to the specific carriers of sensory modalities, all information is subject to modulation and integration by the trigeminal sensory system.

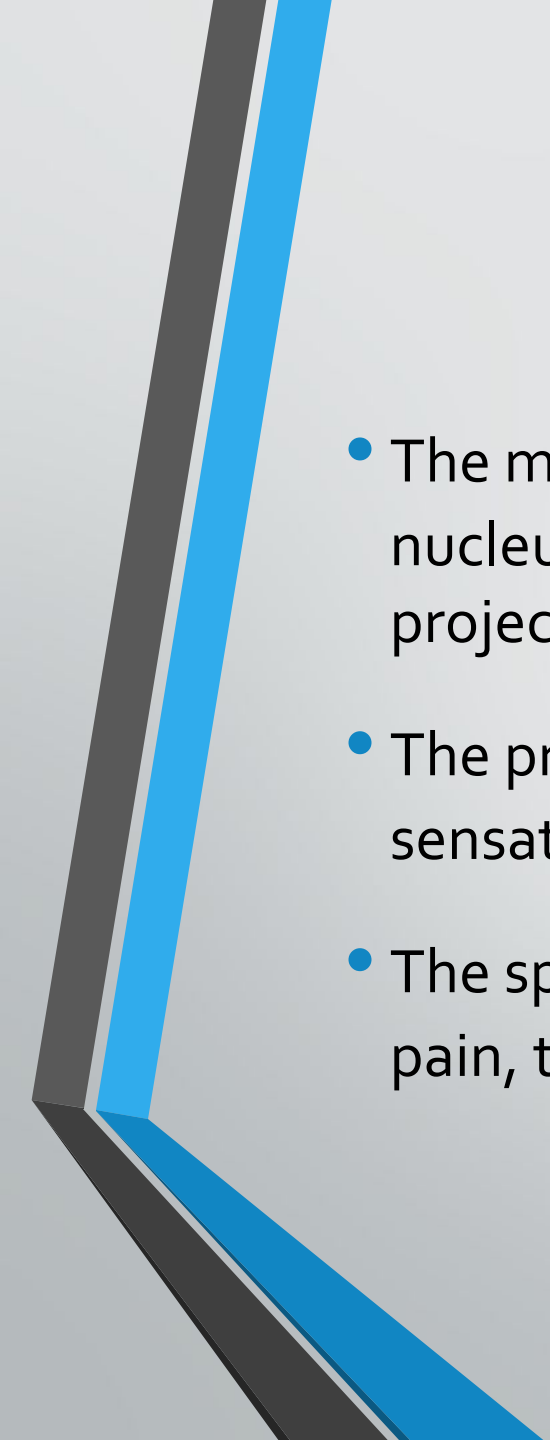
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- The trigeminal nerve is related to four nucleus.
 - Mesencephalic nucleus
 - Main sensory nucleus
 - Spinal nucleus
 - Motor nucleus

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- The sensory trigeminal nerve nuclei are the largest of the cranial nerve nuclei and extend through whole of the brainstem.
 - The trigeminal sensory nuclei are a collection of nuclei located in the brainstem responsible for processing sensory information from the face and head.

The nucleus is divided in three parts

- Mesencephalic trigeminal nucleus
- Principal pontine nucleus
- Spinal nucleus of trigeminal nerve



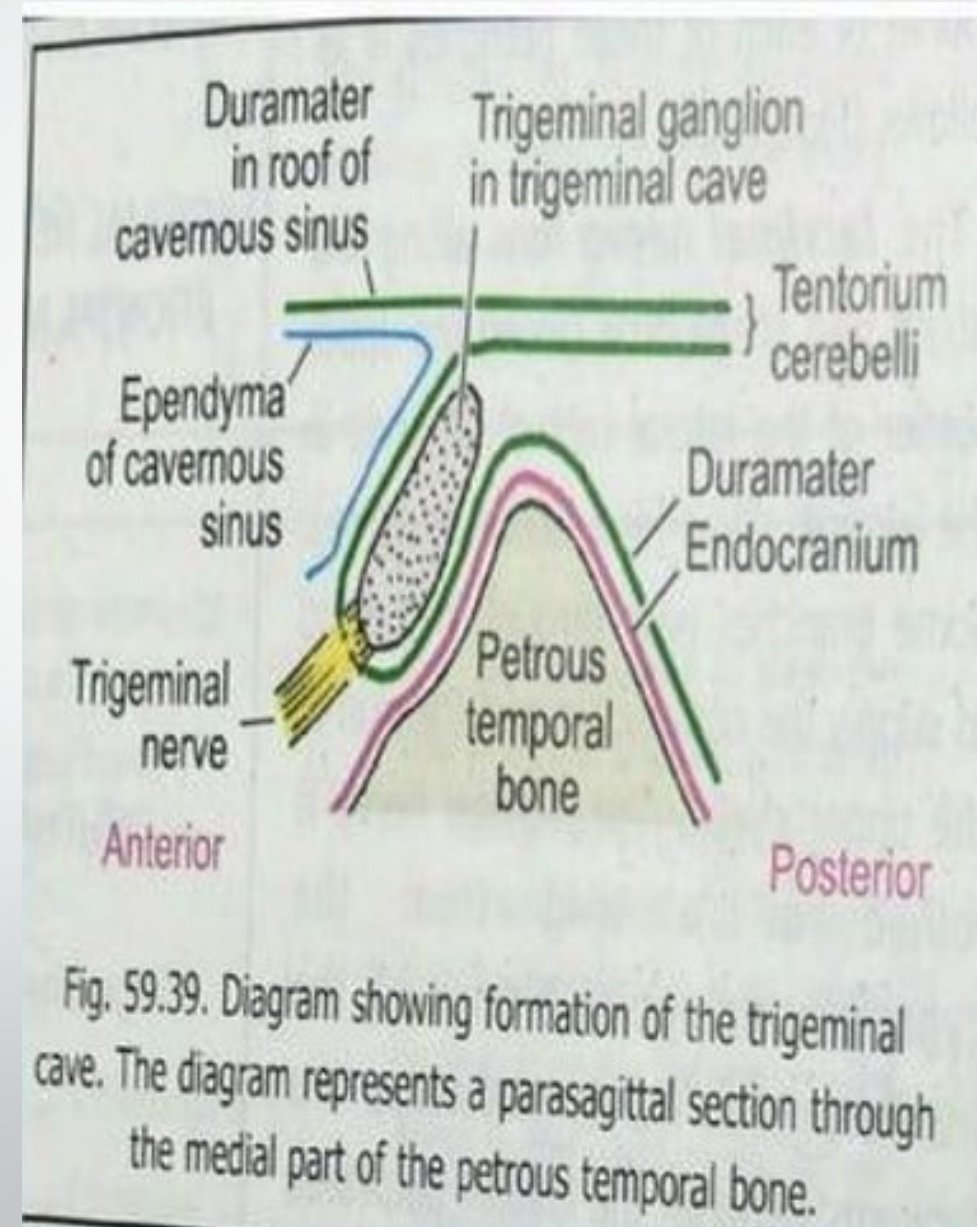
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- The mesencephalic nucleus is involved with proprioception. Neurons of this nucleus cells receiving proprioceptive information from the jaw and send projections to the trigeminal nucleus to mediate jaw reflexes.
 - The principal pontine nucleus receives information about discriminative sensation and light touch of the face.
 - The spinal nucleus of trigeminal nerve receives information about touch, pain, temperature and vibration.

TRIGEMINAL GANGLION

- The trigeminal (semilunar or Gasserian) ganglion is located in the middle cranial fossa at the base of the skull. The neurons within this ganglion receive sensory information via the trigeminal nerve from the skin, mucous membranes, teeth, masticatory muscles (~10%; proprioceptors), Golgi tendon organs, and temporomandibular joint.
- Then, they transmit the information to the brainstem sensory nuclei in the central nervous system (CNS). The sensory submodalities transmitted by the trigeminal ganglion neurons include pain, temperature, pressure, and touch sensations.
- Morphologically, the ganglion neurons are pseudounipolar and are surrounded by satellite cells, which are the protective glial cells in the ganglion.

The Trigeminal Ganglion

- Semilunar or Gasserian ganglion
- Sensory ganglion responding to Dorsal Root Ganglia of spinal nerves.
- Crescentic in shape with convexity anterolat.
- Contains cell bodies of pseudounipolar neurons.
- Location: lies on a bony fossa at apex of the petrous temporal bone on floor of middle



Ganglia Associated with Trigeminal Nerve

- Associated with the three divisions of the trigeminal nerve are four small ganglia.
- The **ciliary ganglion** is connected with the ophthalmic nerve.
- The **sphenopalatine ganglion** with the maxillary nerve.
- And the **otic** and **submaxillary ganglion** with the mandibular nerve.

Divisions

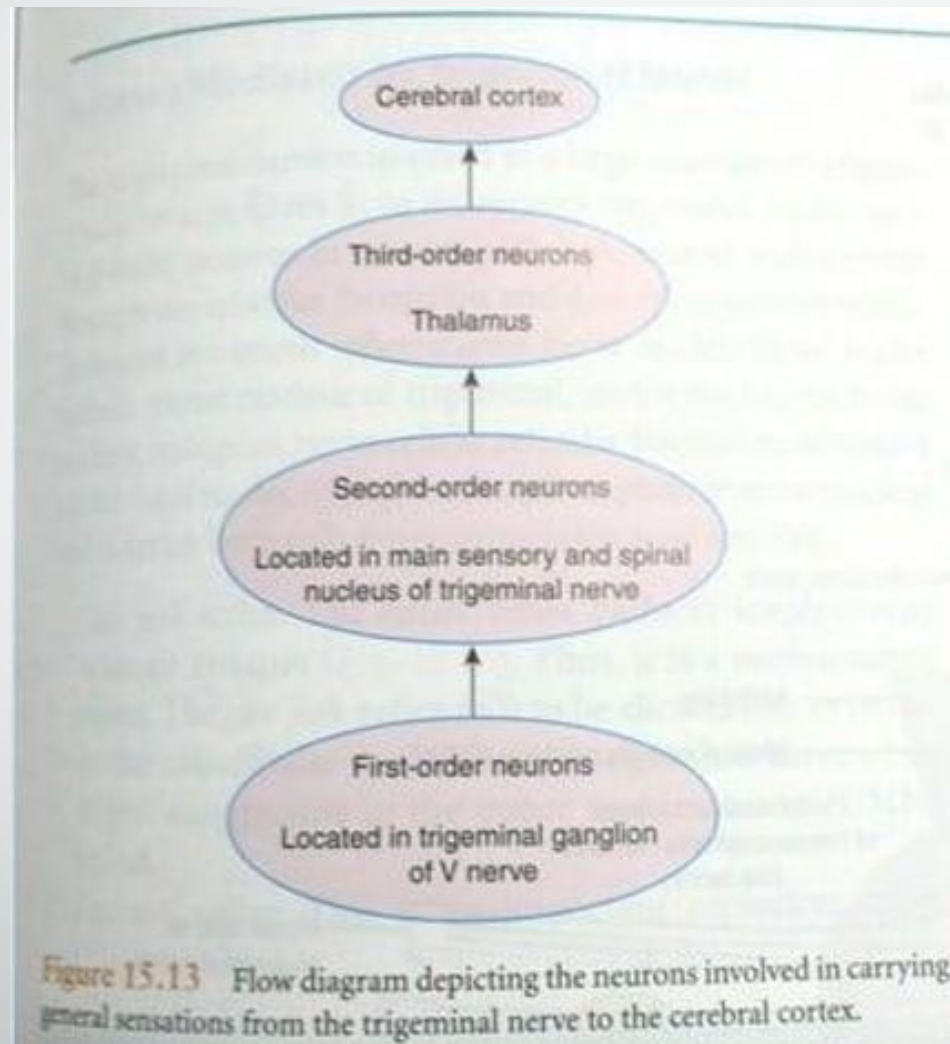
- It has three divisions
 - V₁ - Ophthalmic Division
 - V₂ – Maxillary Division
 - V₃ – Mandibular Division
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- The ophthalmic division enters orbital cavity through superior orbital fissure.
 - The maxillary division exits from cranial cavity through foramen rotundum.
 - The mandibular division passes through foramen ovale



DIVISIONS

- The peripheral nerves originating from the trigeminal ganglion divide into three main branches: ophthalmic (V₁), maxillary (V₂), and mandibular (V₃) branches.
- The V₁ branch exits the skull through the superior orbital fissure, and innervates the forehead, scalp, upper eyelid, and conjunctiva of the eye.
- The V₂ branch exits via the foramen rotundum and traverses the pterygopalatine fossa, providing innervation of the cheek, upper lip, nostril, upper jaw, and teeth.
- The V₃ branch exits through the foramen ovale and the infratemporal fossa, and innervates the lower jaw and teeth, lower lip, cheek, and the anterior two-thirds of the tongue.

General Sensations



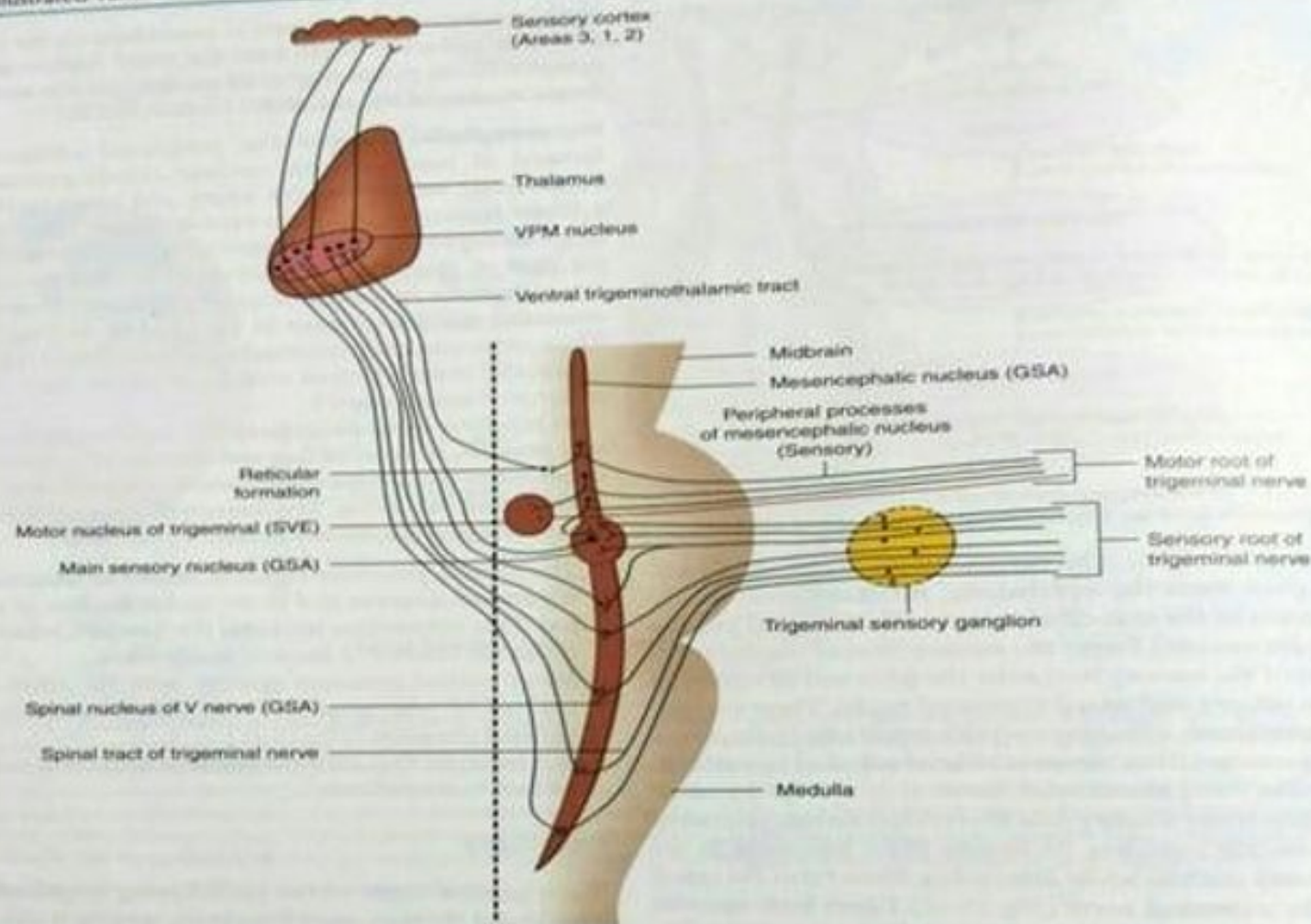


Figure 15.15 Schematic diagram depicting the general somatic afferent column of the V cranial nerve. This column is represented by the mesencephalic, main sensory and spinal nuclei of the trigeminal nerve. The sensory root (central processes) of ganglion cells, which carry the sensation of pain, light touch and temperature after entering the pons, divides into ascending and descending fibres. Ascending fibres terminate in the main sensory nucleus while the descending fibres turn downwards to form the *spinal tract of the trigeminal nerve* in the medulla. The spinal trigeminal tract* descends as far as C2 or C3 segments of the spinal cord. This tract is present just lateral to the spinal trigeminal nucleus. The fibres of the spinal tract terminate upon the neurons of the somatic sensation from the external ear, posterior part of tongue, larynx and pharynx. These fibres bring nucleus and after crossing to the opposite side they ascend close to medial lemniscus of the opposite side. Fibres terminate in the VPM nucleus of thalamus. These fibres form the *ventral trigeminothalamic tract* while fibres arising from the main sensory nucleus are both crossed and uncrossed. They ascend close to central grey and terminate in the VPM nucleus of thalamus. These fibres form the *dorsal trigeminothalamic tract* (not shown in this figure). GSA—general somatic afferent; SVE—special visceral efferent; VPM—ventral posteromedial nucleus.

SENSORY ROOTS

GENERAL SOMATIC AFFERENTS

Face, Scalp, Teeth, Gingiva, Oral, Nasal Cavity, PNS, Conjunctive and Cornea

Pain, temp, light touch

Touch pressure

Proprioception

Trigeminal gang.

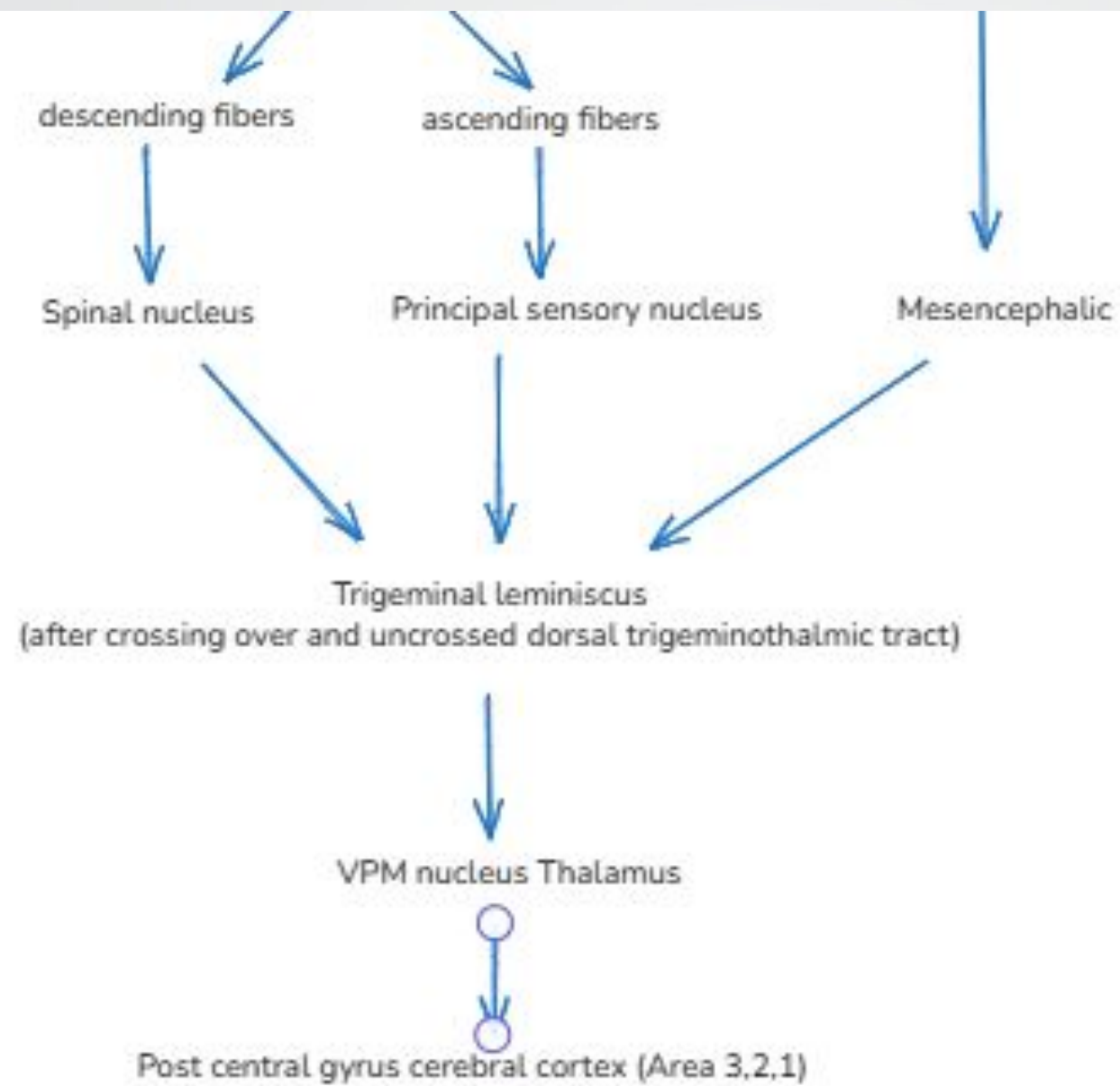
Bypass trigeminal gang.

sensory root

descending fibers

ascending fibers





Motor nucleus



Motor root



Mandibular nerve



muscles of mastication

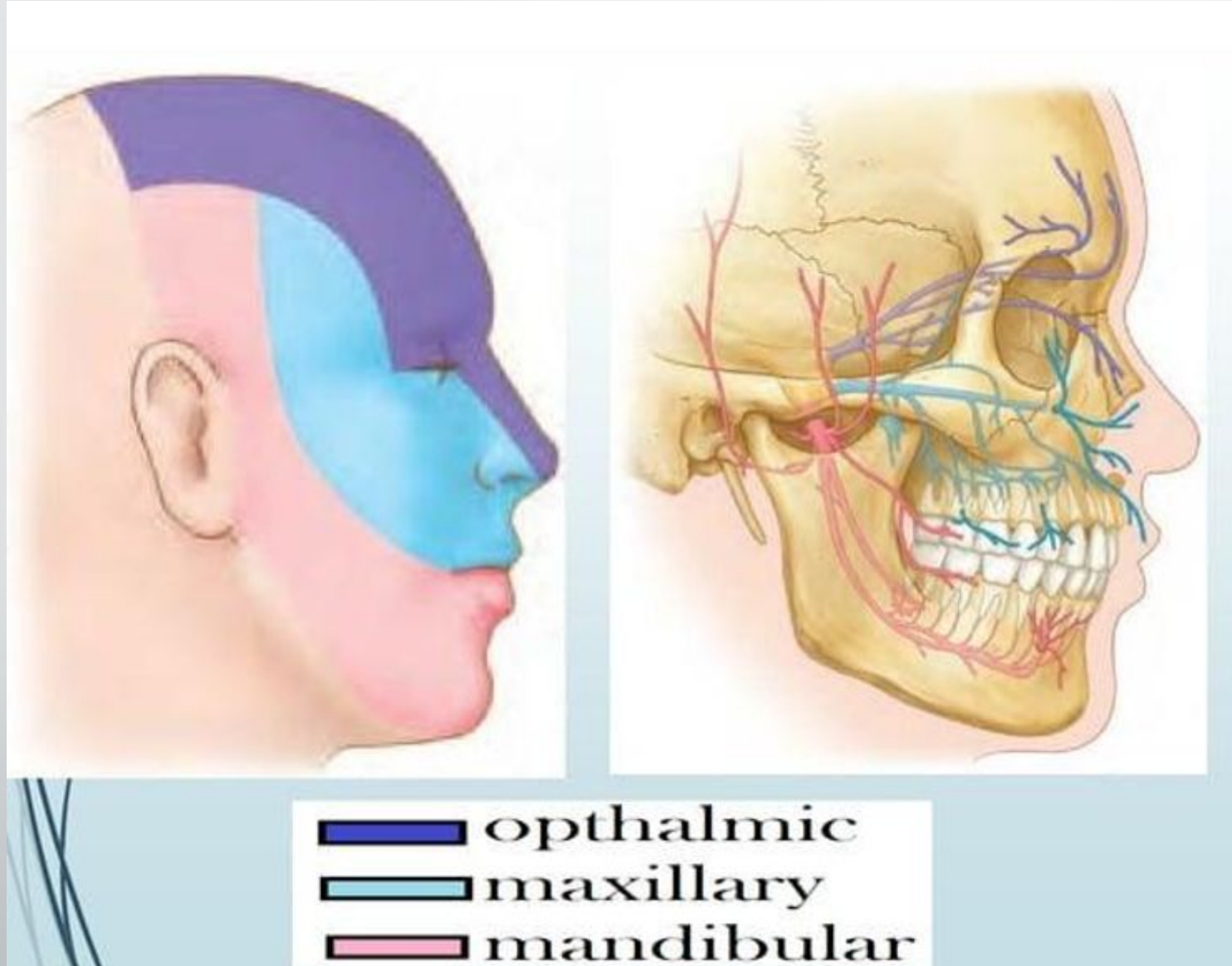
tensor tympani

masseter

tensor palatini

lat / med pterygoids

temporalis



Ophthalmic Nerve

- Smallest of the three branches.
- Purely sensory
- Arises from the anteriomedial end of the semilunar ganglion and passes forward in the lateral wall of cavernous sinus.
- As ophthalmic division passes forward from cavernous sinus, it divides into three branches.
- Lacrimal
- Frontal
- Nasociliary

emerges from trigeminal ganglion

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graph TD; A[emerges from trigeminal ganglion] --> B[lateral wall cavernous sinus]; B --> C[3 branches in ant part of cavernous sinus, lacrimal, nasocilliary, frontal.]; C --> D[superior orbital fissure]; D --> E[orbit];
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lateral wall cavernous sinus

3 branches in ant part of cavernous sinus, lacrimal, nasocilliary, frontal.

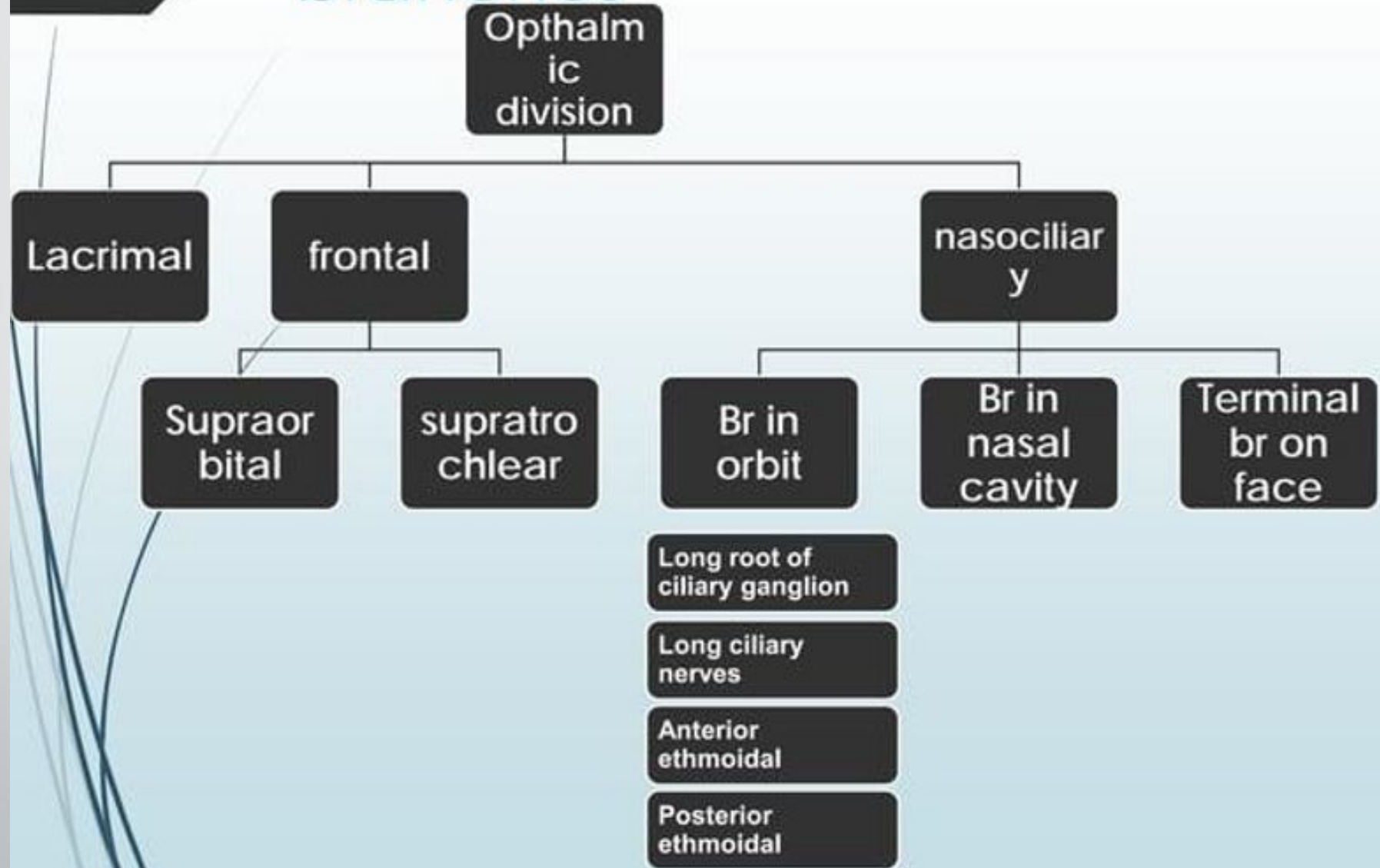
superior orbital fissure

orbit

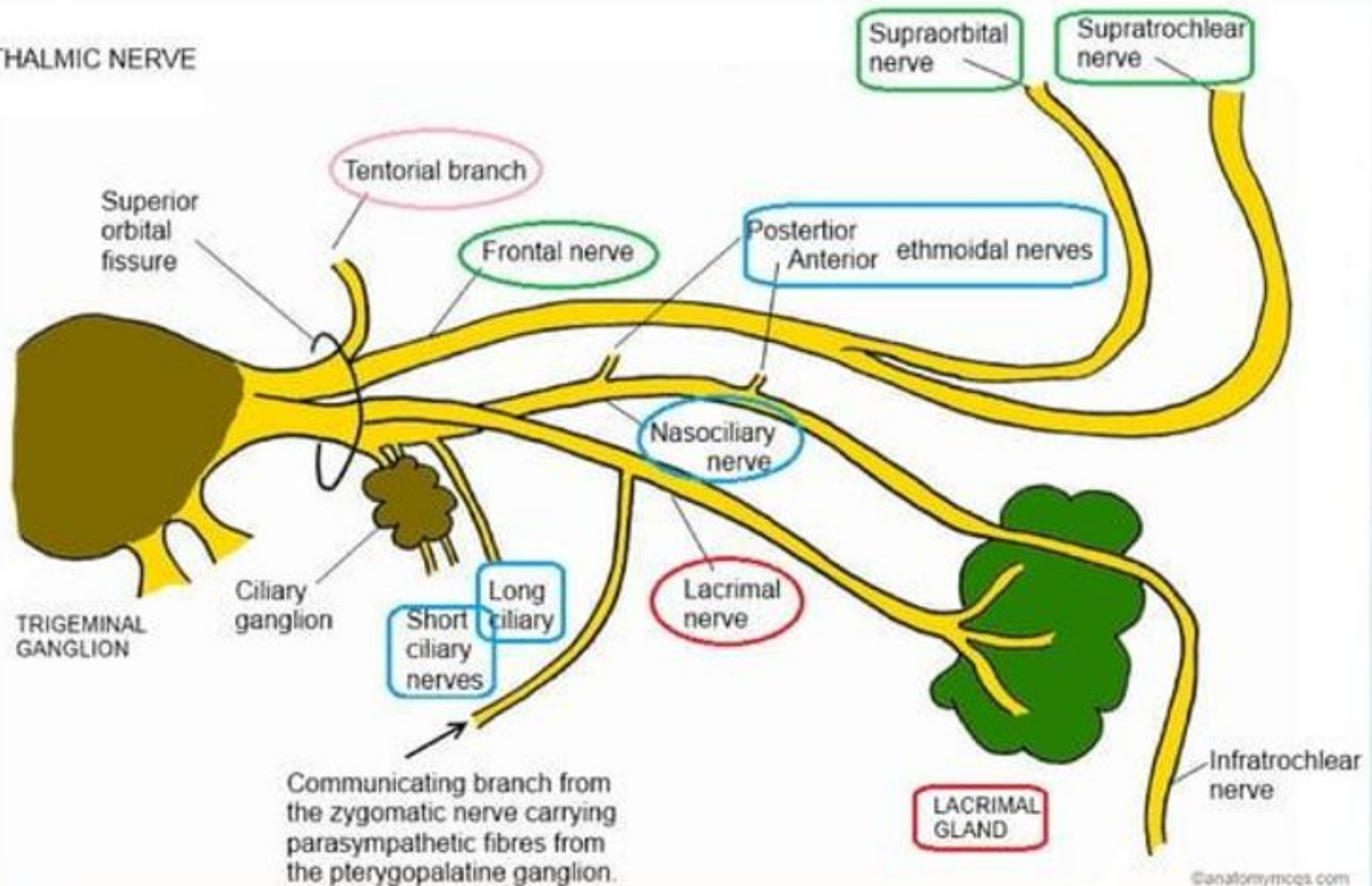
Sensory or Afferent Fibers from:

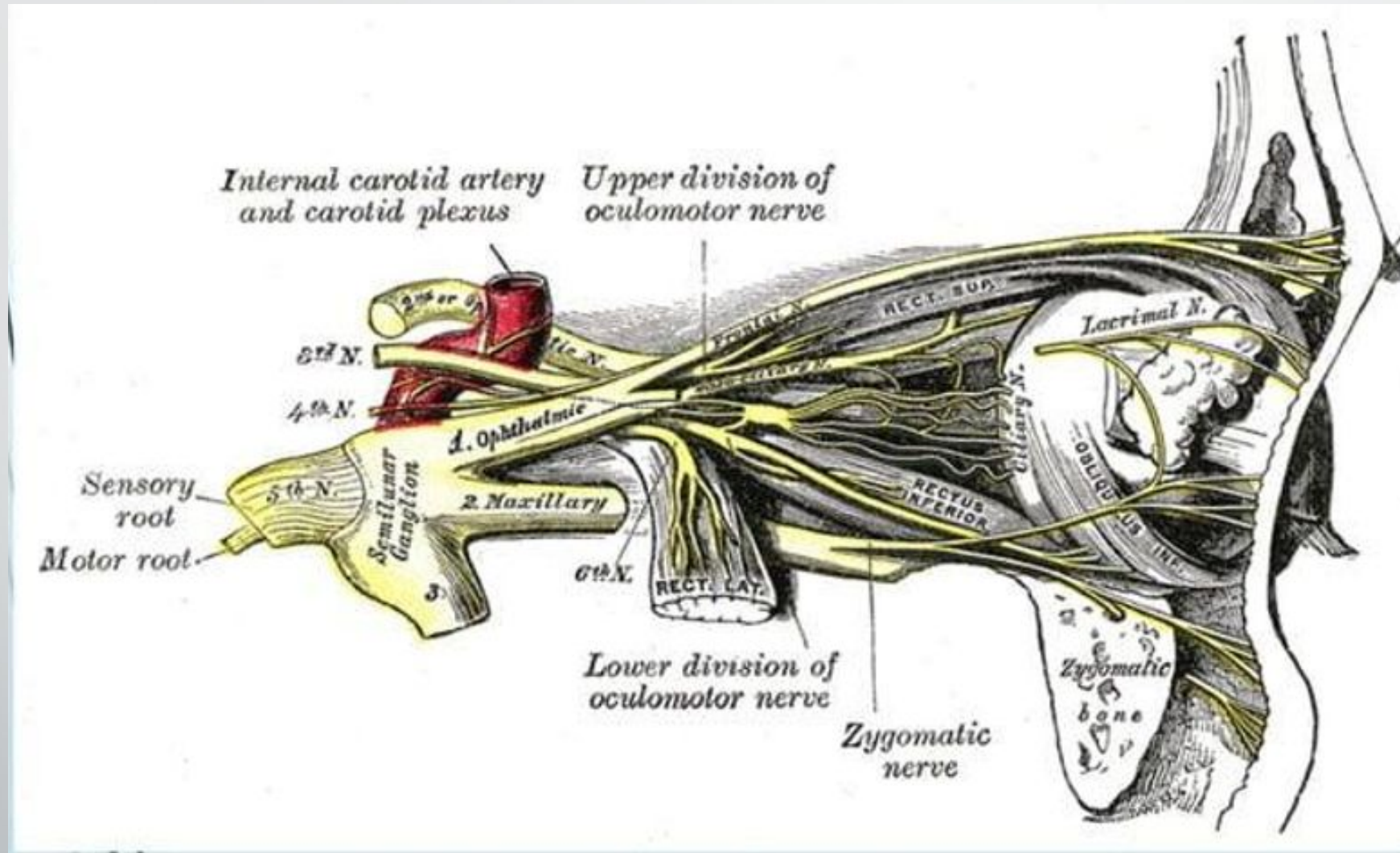
- Scalp
- Skin of forehead
- Conjunctiva
- Sclera
- Lacrimal gland
- Skin of the lateral angle of eyeball
- Lining of the ethmoidal cells
- Upper eyelid lining the frontal sinus

branches



OPHTHALMIC NERVE





Lacrimal Nerve

- smallest of the three branches
- Passes into orbit at the lateral angle of SOF
- It runs forward on the upper border of the lateral rectus muscle. It is joined by the zygomaticotemporal branch of the maxillary nerve, which contains the parasympathetic secretomotor fibers to the lacrimal gland.
- The lacrimal nerve then enters the lacrimal gland and gives branches to the conjunctiva and the skin of the upper eyelid.
- Occasionally it is absent, in which case it is replaced by the zygomaticotemporal nerve: the relationship is reciprocal.
- Receives communicating branch from trochlear nerve

Corneal Reflexes

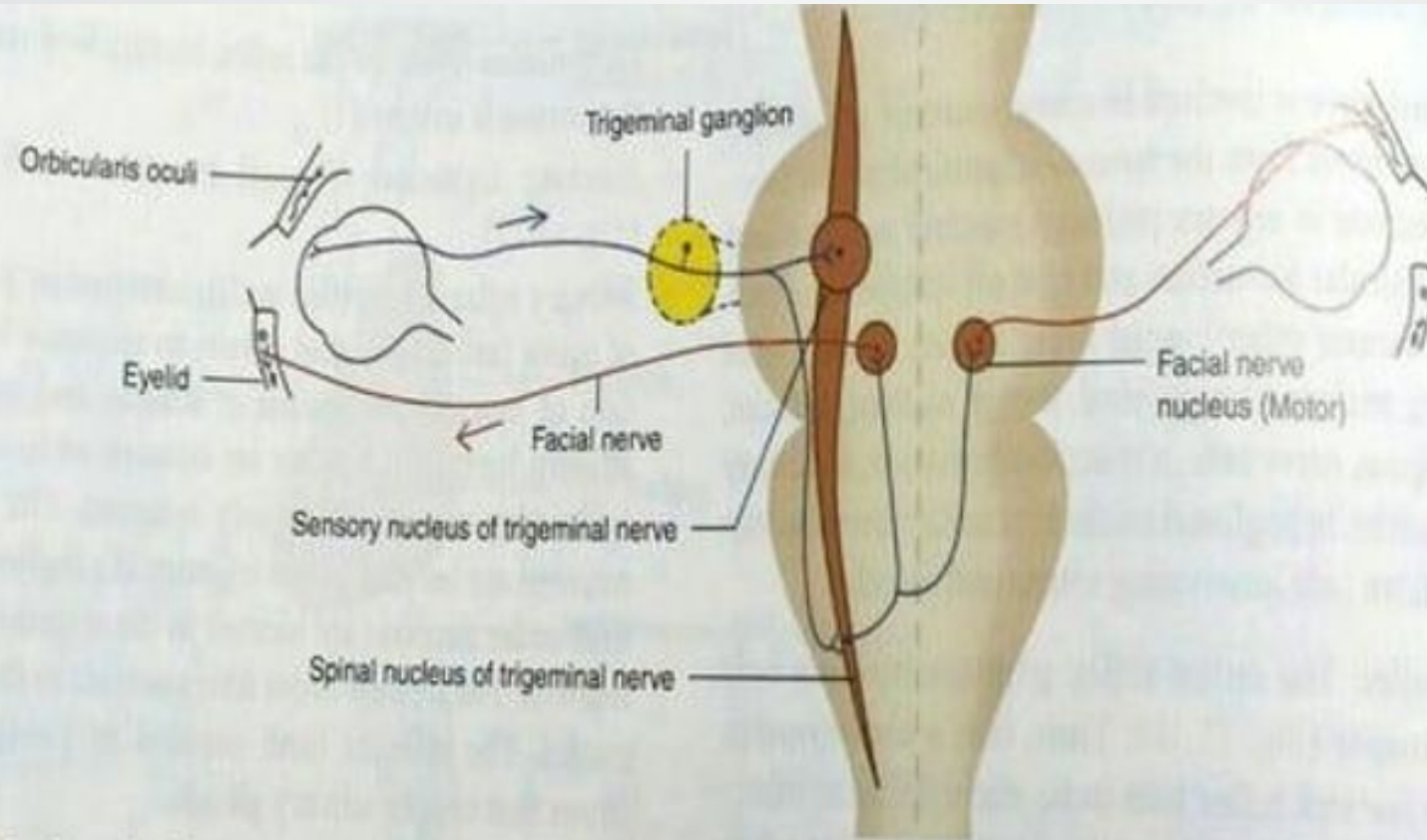


Figure 15.17 Corneal reflex: Reflex closure of eyelids (on both sides) if the cornea is touched with a wisp of cotton. The afferent impulses from the cornea are carried through the ophthalmic division. The cell bodies of these afferent fibres are situated in trigeminal ganglion (*first-order neurons*). The second-order afferent neurons are situated in trigeminal sensory nuclei. These impulses reach the facial nuclei on both sides. The efferent fibres from the facial nuclei project to the eyelids bilaterally, leading to bilateral closure of eyelids. The fibres of second-order neurons project to facial motor nuclei of both the sides. The efferent fibres of the reflex arc consist of facial nerve fibres that innervate orbicularis oculi muscle.

Clinical Application of Trigeminal Ganglion

- Shingles and varicella-zoster: The trigeminal ganglion, as any sensory ganglion, may be the site of infection by the herpes zoster virus causing shingles, a painful vesicular eruption in the sensory distribution of the nerve.
- Trigeminal neuralgia (tic douloureux): This is severe pain in the distribution of the trigeminal nerve or one of its branches, the cause often being unknown. It may require partial destruction of the ganglion.

Clinical Application

- **Ethmoid tumours:**

Malignant tumours of the mucous lining of the ethmoid air cells may expand into the orbits, damaging branches of ophthalmic nerve. This may lead to displacement of the orbital contents causing proptosis and squint, and sensory loss over the anterior nasal skin.

- **Nasal fractures:**

Trauma to the nose may damage the nasociliary nerve. Sensory loss of the skin down to the tip of the nose may result.

- **Corneal Reflex:**

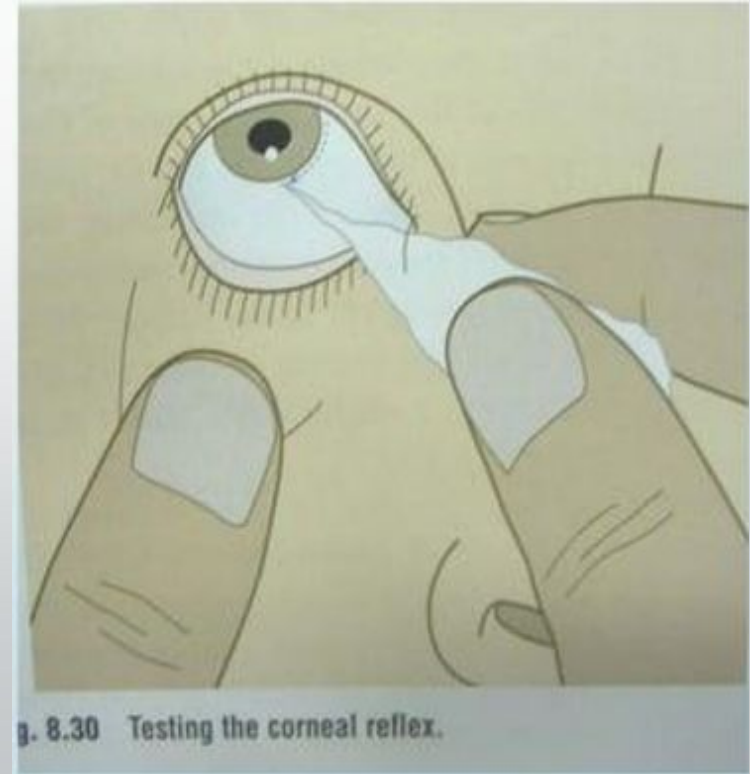
When the cornea is touched, usually with a wisp of cotton, the subject blinks. This test V and VII. The nerve impulses pass through cornea, and then through nasociliary nerve to the brain.

- **Supraorbital Injuries:**

Trauma to the supraorbital margin may damage the supraorbital and supratrochlear nerves causing sensory loss in the scalp.

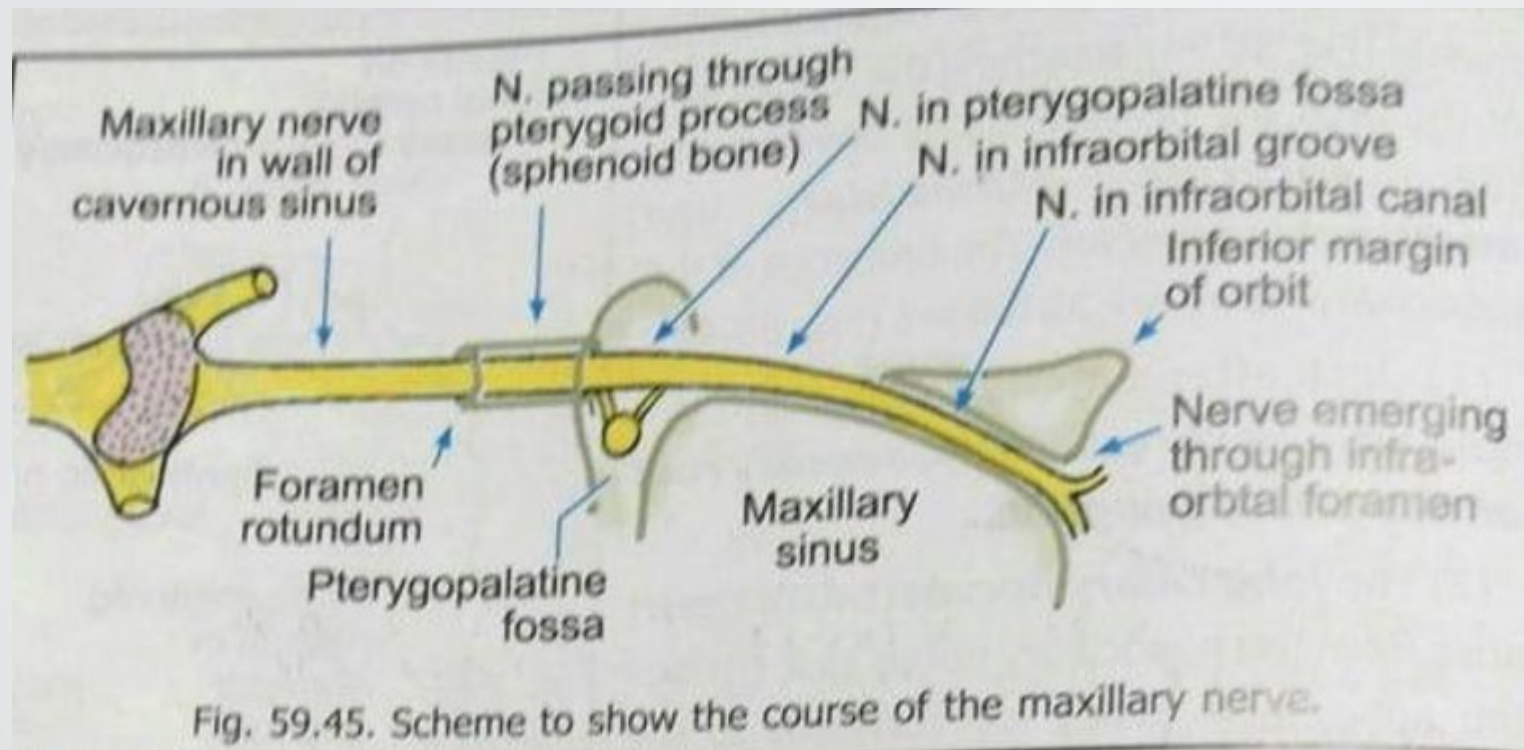
Corneal Reflex

- Ask the patient to look upward to the ceiling and gently depress the lower eyelid
- Lightly touch the lateral edge of the cornea with damp cotton wool
- Look for both direct and consensual blinking



Maxillary Nerve

- Second division of trigeminal nerve
- Purely sensory
- Supplies derivatives of maxillary process and frontonasal process.



Course: trigeminal gang. Middle cranial fossa
 lat wall of cavernous sinus
 foramen rotundum
 pterigopalatine fossa
 in groove on post surf of maxilla
 through inf orbital fissure into orbit as INFRA ORBITAL N
 through infraorbital foramen on face

Innervations

- **Skin:**
 - Middle portion of face
 - Lower eyelid
 - Side of nose
 - Upper lip
- **Mucous Membrane:**
 - Nasopharynx
 - Maxillary sinus
 - Soft palate
 - Hard palate
 - Tonsile
 - **Maxillary teeth and pdl**

Branches of Maxillary Nerve

It is divided into four groups as they are given off:

- Middle Cranial Fossa
- Pterygopalatine Fossa
- Infraorbital Canal and Groove
- On the Face

MIDDLE CRANIUM FOSSA	MIDDLE MENINGEAL NERVE
PTERYGOPALATINE FOSSA	ZYGOMATIC
	SPHENOPLATINE
	POSTERIOR SUPERIOR ALVEOLAR
INFRAORBITAL CANAL	ANTERIOR SUPERIOR ALVEOLAR
	MIDDLE SUPERIOR ALVEOLAR
ON THE FACE	INFERIOR PALPERBAL
	EXTERNAL NASAL
	SUPERIOR LABIAL

MAXILLARY NERVE

Within cranium

Middle
meningeal
nerve

In pterygopalatine fossa

Ganglionic
branch

Zygomatic

Zygomatico
temporal

Zygomatico
facial

Pterygopalatine

Orbital
Nasal
Palatine
Pharyngeal

Orbital
Nasal
Palatine

PSA
(posterior
superior
alveolar)

In infraorbital canal

MSA
(middle superior
alveolar nerve)

ASA
(anterior superior
alveolar nerve)

On face

➤ Inferior
palpebral

➤ Lateral nasal

➤ Superior labial

Within Cranium

Middle meningeal nerve

- It leaves the **maxillary nerve** near the **foramen rotundum**.
- It runs along with the **middle meningeal artery** to supply the **dura mater** in the **middle cranial fossa**.

In Pterygopalatine Fossa

1. Ganglionic branches

- Related to pterygopalatine ganglion
- Carry sensations from
 - Orbital periosteum
 - Nose
 - Pharynx
 - Palate
- Carry post-ganglionic parasympathetic secretomotor fibres to the **lacrimal gland**.

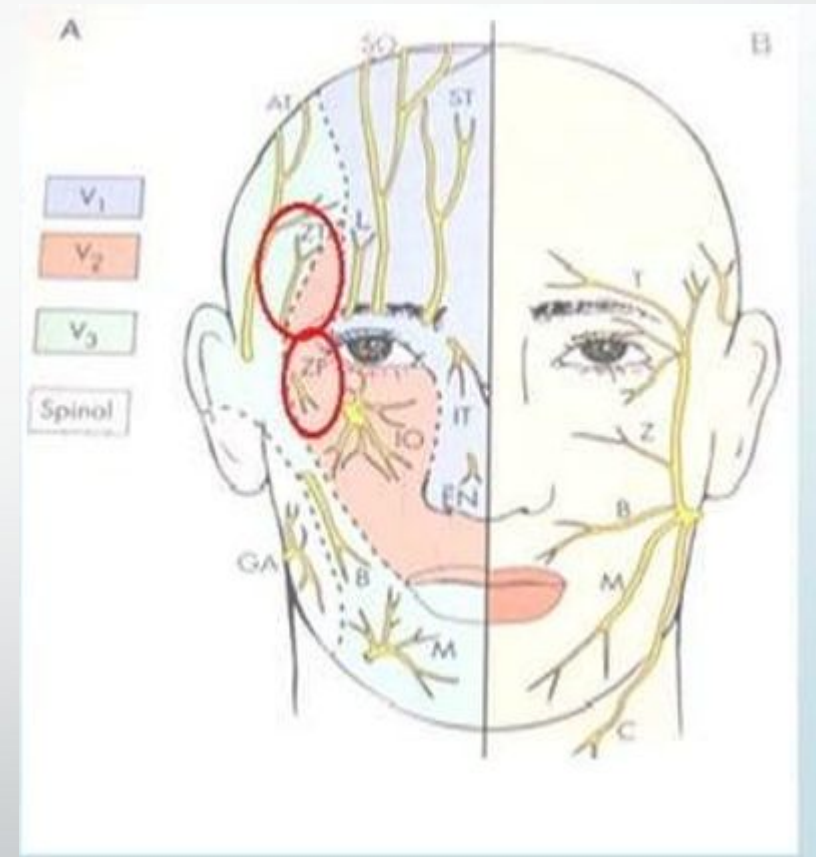
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- **Zygomatic Nerve:**
 - Starts in the pterygopalatine fossa.
 - Enters the orbit through the infraorbital fissure along its lateral border, where it divides into 2 branches:
 - **Zygomaticotemporal**
 - **Zygomaticfacial**

A. Zygomaticofacial nerve

- **Appears on face** through a foramen in the zygomatic bone.
- **Supplies skin** on the prominence of the **cheek**.

B. Zygomaticotemporal nerve

- **Supplies skin** of the **temporal region** after piercing the temporal fascia **2 cm above the zygoma**.
- **Gives a communicating branch** to the lacrimal nerve.



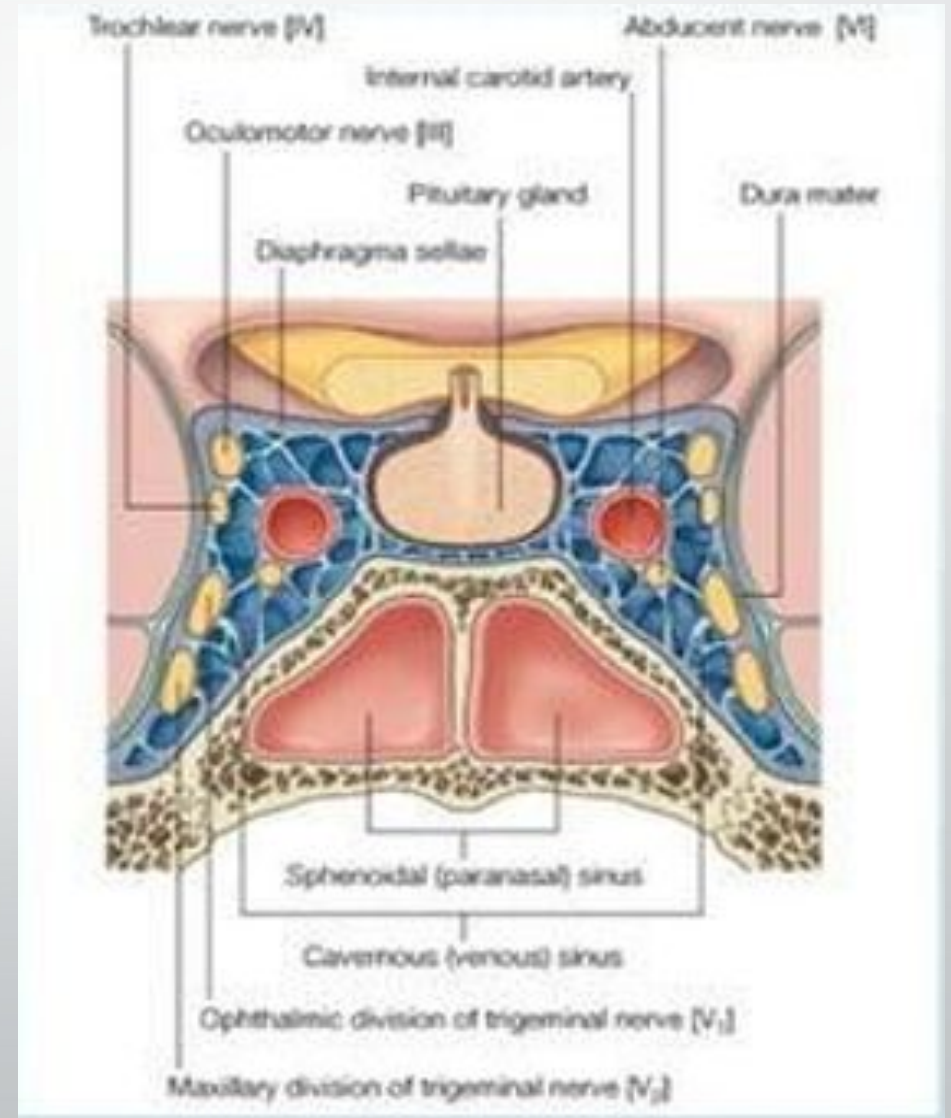
Applied Anatomy

Causes of injury to Maxillary nerve

- Maxillofacial surgical procedures
 - Orthognathic surgeries
 - Head & neck preprosthetic surgeries
 - Treatment of benign & malignant lesions
- Trauma & facial fractures
- Dental implant placement
- Endodontic therapy
- Treatment of pathology (especially periapical)
- During administration of local anesthesia

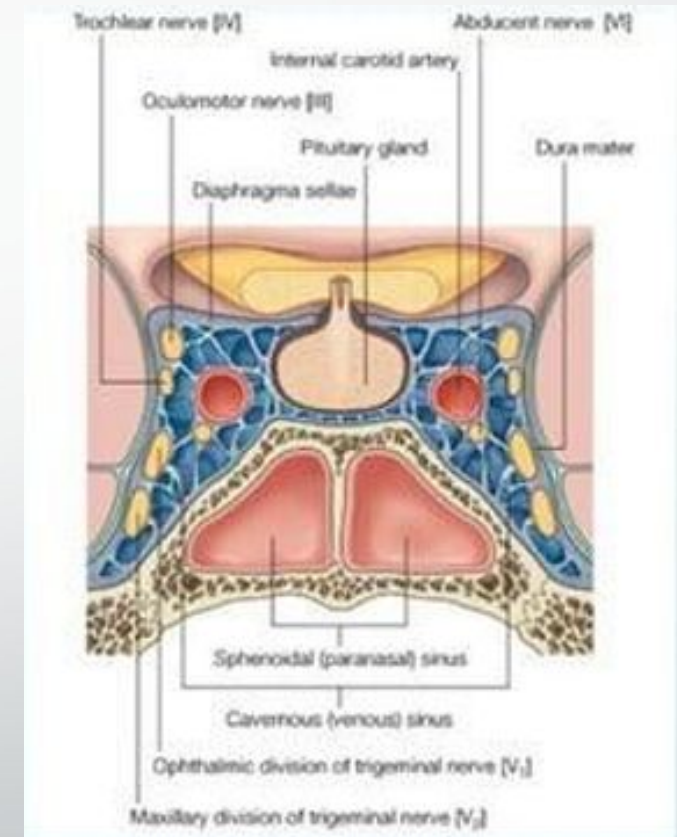
Cavernous Sinus Thrombosis

- Cavernous sinus syndrome is a medical emergency, requiring prompt medical attention, diagnosis, and treatment.
- Result from involvement of CS (Cavernous Sinus) by inflammatory/septic foci.



Potential causes of cavernous sinus syndrome include:

- Metastatic tumors
- Direct extension of nasopharyngeal tumors
- Meningioma
- Pituitary tumors
- Aneurysms of the intracavernous carotid artery
- Bacterial infection causing cavernous sinus thrombosis
- Aseptic thrombosis
- Fungal infections



Clinical Features

- High grade fever
- Altered consciousness
- Severe infection in danger area of face
- Chemosis
- Proptosis
- Ophthalmoplegia

Multiple cranial neuropathies

- Exophthalmos
- Sensory loss in V1 and/or V2

Treatment

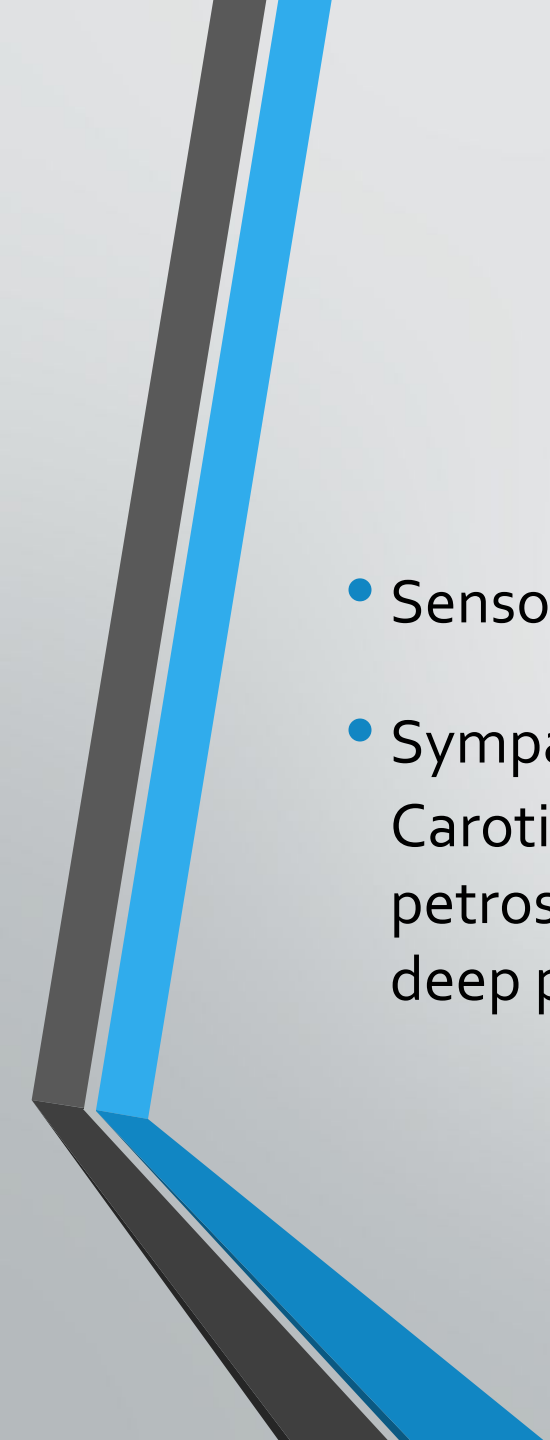
- **Early and aggressive Broad spectrum parenteral antibiotic administration** for 3–4 weeks.
- **IV mannitol** to reduce intracranial pressure.
- **Anticoagulant** to prevent ext. (extension) of thrombosis.
Heparin: 20,000 units in 1,500 ml of D5
OR
Dicumarol: 200mg orally followed by 100mg daily.
- **Corticosteroid:** reduced intracranial tension, but there is a risk of spread of infection.
- **Neurosurgical intervention** is mandatory.

Maxillary Sinus Infections

- Infections of the maxillary sinus may cause infraorbital pain or may cause referred pain to other structures supplied by Vb (eg: Upper teeth)

Pterygopalatine Ganglion

- **Sphenopalatine ganglion** is the largest parasympathetic ganglion, suspended by two roots of **maxillary nerve**.
- **Functionally** it is related to the **facial nerve**.
- It is also the **ganglion of hay fever**.
- **Roots:**
 - **Sensory, sympathetic and secretomotor or parasympathetic roots**

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- Sensory roots is from maxillary nerve.
 - Sympathetic roots is from postganglionic plexus around ICA (Internal Carotid Artery).The nerve is called deep petrosal. It unites with greater petrosal to form the nerve of pterygoid canal (Vidian nerve).The fibres of deep petrosal nerve do not relay on ganglion.

Clinical Examination of Maxillary Nerve

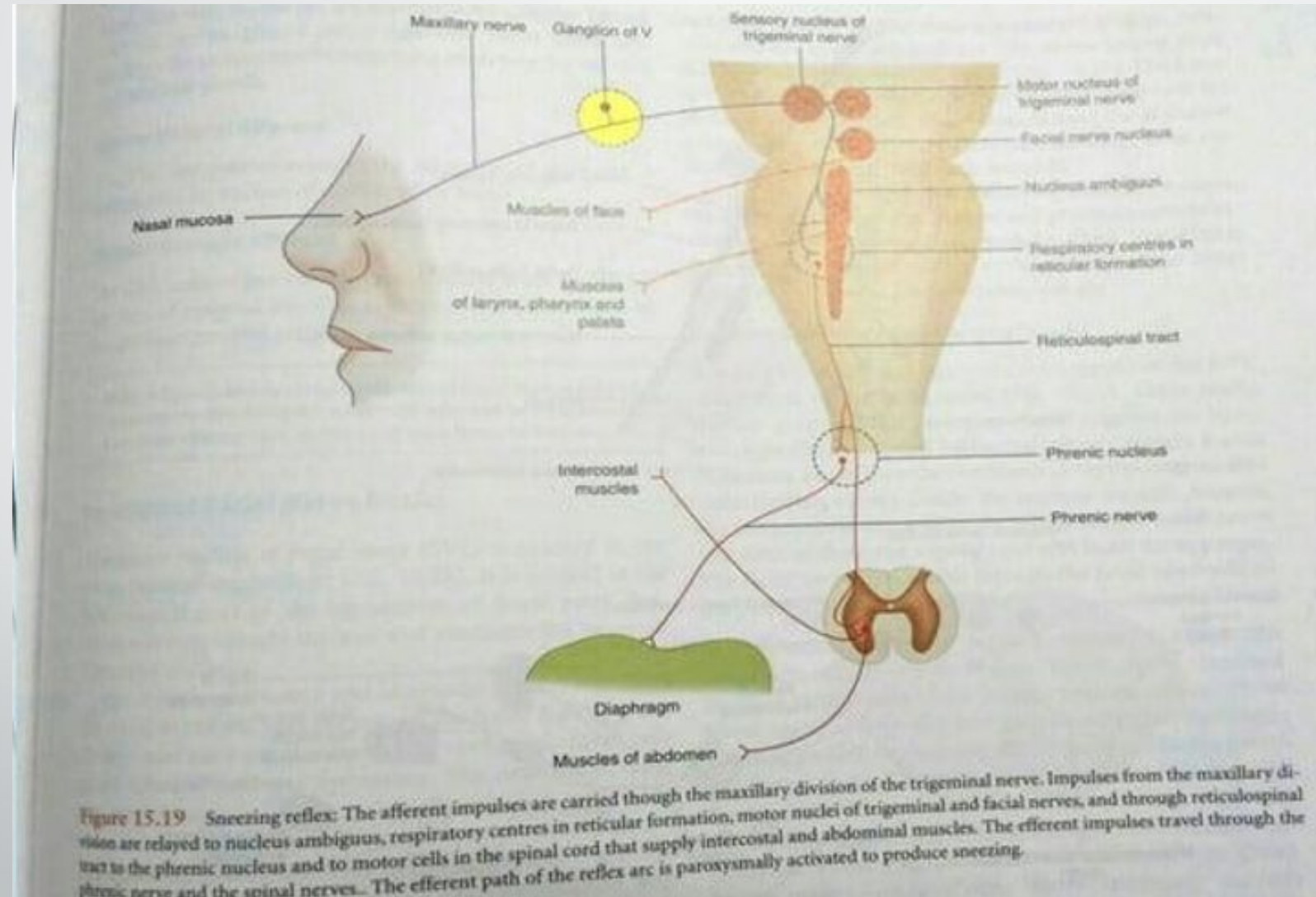
- Sensory:

Apply gentle touch, painpricks, or warm or cold objects to areas supplied by the nerve and and note responses.

- Reflex:

Sneeze reflex

Sneeze Reflex



Mandibular Nerve

- Mandibular nerve is the largest branch of the trigeminal nerve.
- Mixed nerve with two roots:
- Large sensory: from inferior angle of TGG (Trigeminal Ganglion)
- Small motor: Motor cells located in pons & medulla

Innervations

1. Sensory roots:

a) Skin:

- Temporal region
- Auricula
- Ext. auditory meatus
- Cheek
- Lower lip
- Lower part of face

b) Mucous membrane:

- Cheek
- Tongue
- Mastoid cells

c) Mandibular teeth and pdl (periodontal ligament)

d) Bone of mandible

e) TMJ (Temporomandibular joint)

f) Parotid gland



2. Motor roots

a. Masticatory muscles:

- Masseter
- Temporalis
- Medial pterygoid
- Lateral pterygoid

b. Mylohyoid

c. Ant. Belly of digastric (Anterior belly of the digastric muscle)

d. Tensor tympani

e. Tensor veli palatini

Sphenopalatine Neuralgia

- ETIOLOGY: Vasodilatation of the internal maxillary artery near the region of the Sphenopalatine region.
- Unilateral paroxysms of intersparse pain in the region of eyes, maxilla, ear, mastoid, base of the nose and beneath the zygoma.
- There are no TRIGGER ZONES.
- Seen in persons below 40 years.
- Treatment:

Alcohol injection of the ganglion

Resection of the ganglion

Auriculotemporal Syndrome

ETIOLOGY

- It results from damage to the **Auriculotemporal nerve**.
- The syndrome usually follows some **surgical operation** (most commonly parotidectomy).

CLINICAL FEATURES

- **Flushing and sweating** of the involved side of the face (specifically during eating).

TREATMENT

- **Intracranial division** of the auriculotemporal nerve.

Orofacial Pain Related to the Trigeminal Sensory System

1. Orofacial Neuropathic Pain; The trigeminal nerve, which is responsible for sensation in the mouth, teeth, and face, plays a crucial role in the perception of pain in the orofacial region.

Injury or inflammation of this nerve can result in chronic neuropathic pain, characterized by spontaneous pain, hyperalgesia, and allodynia.

The underlying mechanisms contributing to the development and maintenance of orofacial neuropathic pain include peripheral and central sensitization, the involvement of ion channels and neurotransmitters, and neuroinflammation .

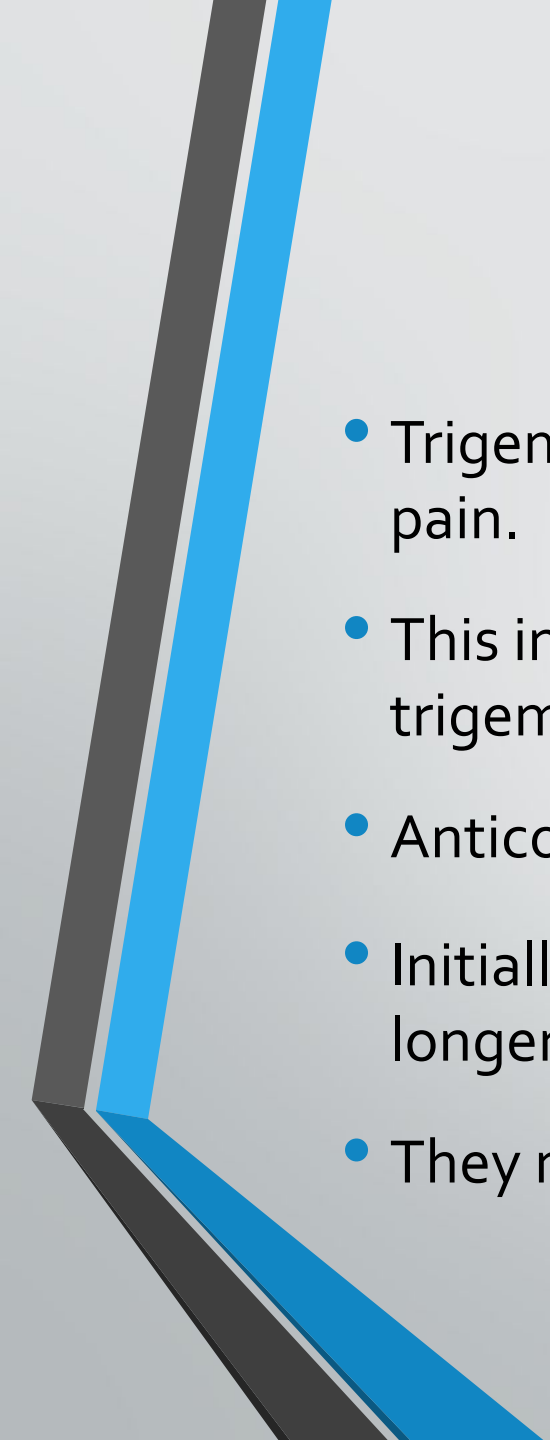
- **Peripheral sensitization** plays a crucial role in the initiation of orofacial neuropathic pain
- . Following injury or damage to the trigeminal nerve, primary afferents, or nociceptors in the affected area, become hyperexcitable . This hyperexcitability results from various changes at the molecular and cellular levels, including the upregulation of ion channels, alterations in receptor function, and increased expression of pro-inflammatory mediators .
- After nerve injury, there is an increased expression of voltage-gated sodium channels (VGSCs) and VGCCs in the damaged neurons .
- This upregulation leads to enhanced excitability and spontaneous ectopic discharges, contributing to the development of neuropathic pain triggered by nerve injury . The function of receptors such as TRP channels, including TRPV₁, which is involved in heat sensation and pain, is altered, resulting in heightened sensitivity to thermal and mechanical stimuli
- Additionally, injury to the trigeminal nerve leads to the release of pro-inflammatory cytokines (e.g., TNF- α , IL-1 β , IL-6) and chemokines, which sensitize nociceptors and contribute to the development of peripheral sensitization.

- **Central Sensitization** involves hyperexcitability of central neurons, LTP, microglial activation, and a loss of inhibitory control
- Following peripheral nerve injury, the hyperexcitability of neurons in the spinal trigeminal nucleus and higher centers, such as the thalamus and cortex, is attributed to be caused by an increased release of excitatory neurotransmitters, including glutamate and substance P, alongside a decrease in inhibitory neurotransmission. Central sensitization is linked to LTP of synaptic connections and is also observed in cases of orofacial neuropathic pain.
- The upregulation of VGSCs in injured neurons increases neuronal excitability, and mutations in these channels, particularly in Nav1.7, have been linked to neuropathic pain disorders
- Neuroinflammation in orofacial neuropathic pain is closely associated with both peripheral and central neuroinflammation, which contribute to the pathogenesis of this condition.

Peripheral neuroinflammation involves the release of pro-inflammatory cytokines and chemokines from immune cells, glial cells, and neurons, creating a pro-inflammatory environment that sensitizes peripheral nociceptors and generates pain signals. Central neuroinflammation occurs when microglia and astrocytes in the CNS become activated in response to peripheral nerve injury. Activated glial cells release cytokines, chemokines, and other mediators that promote neuronal hyperexcitability and sustain central sensitization.

Trigeminal Neuralgia (Tic Douloureux)

- Trigeminal neuralgia (TN) is defined by the International Association for the Study of Pain as “sudden, usually unilateral, severe, brief, stabbing, recurrent pains in the distribution of one of more branches of the fifth cranial nerve”, and is particularly triggered by innocuous stimuli affecting the face or intraoral trigeminal territory.
- Classic or typical TN can be classified as either idiopathic (primary; without identifiable structural lesions) or secondary to lesions such as tumors , cysts, or multiple sclerosis.
- TN is caused by compression of the trigeminal roots by small, tortuous branches of the basilar artery, primarily the superior cerebellar artery and, occasionally, the anterior inferior cerebellar artery, in the posterior fossa.
- Both vascular compression and conditions like multiple sclerosis and tumors lead to focal demyelination and injury of the trigeminal nerve , resulting in hyperexcitable axons, ectopic generation of impulses with high-frequency afterdischarges, and ephaptic (extrasynaptic) transmission of electrical impulses from low-threshold touch fibers to pain fibers due to the transmembrane passage of ions between axons .

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- Trigeminal neuralgia or tic douloureux is described as the most excruciating pain.
 - This intense, stabbing, electric shock-like pain is caused by irritation of the trigeminal nerve.
 - Anticonvulsive medications are normally the first treatment choice.
 - Initially short, mild attacks, but trigeminal neuralgia can progress, causing longer, more frequent bouts of searing pain.
 - They may also be provoked by even mild stimulation of the face.

Precipitating Factors

- Shaving
- Stroking your face
- Eating
- Drinking
- Brushing your teeth
- Talking
- Putting on makeup
- Encountering a breeze
- Smiling